Rebecca Lee

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Education

Tufts University Medford, MA

B.S. Computer Science, May 2027 | M.S. Human-Robot Interaction (5th-Year), Expected May 2028 GPA: 3.74/4.0, Dean's List

- **Relevant Coursework:** Game Design, Algorithms, Programming Languages, Computer Architecture, Database Systems

Experience

Data Management and Automation Intern June 2025 – October 2026

Orna Therapeutics *Watertown, MA*

- Automated processing of research data files with Python and Pandas, replacing manual data entry for the research team
- Designed and deployed SQL dashboards visualizing test results for cross-functional teams, informing research priorities
- Partnered with scientists and engineers in agile sprints to gather requirements and iterate on database query optimization

Owlhead VR Programming Team Lead September 2024 – Present

Tufts University Idealab *Medford, MA*

- Lead a 6-person team building a Unity (C#) VR experience that widens the wearer's horizontal vision to ∼180° by bringing 3 live camera feeds (fisheye + 2 USB) into the headset; run sprints and assign tasks
- Integrated a bidirectional treadmill for controller-free locomotion: physical walking drives in-game movement, with turning implemented despite the single-axis treadmill; used to play games such as Minecraft
- Developing a custom C# stitching script to replace the current OBS-based stitching stage; next up are smoother turning, treadmill safety, and in-game jumping
- Presented the team's research at the Microsoft NERD Center in Boston (April 2025)

Future Educational Technologies Lab Research Assistant September 2025 – Present

Tufts Center for Engineering Education Outreach *Medford, MA*

- Architected an IoT-enabled React web app connecting ESP32C6 microcontrollers to teachers' mobile devices for Smart Playground robotics activities, supporting real-time control of 20+ concurrent robot connections
- Engineered a Python communication layer over Bluetooth and WiFi for ESP32C6 devices using asynchronous event handling
- Modeled a crank-driven LEGO quadruped in MuJoCo and trained walking policies with PPO (Stable Baselines3); wrote calibration scripts to deploy policies to the physical robot (sim-to-real)

Jumbo Code Tech Lead, DevOps October 2026 – Present

Tufts University *Medford, MA*

- Lead a team of 3 developers maintaining and extending past nonprofit client projects: write and assign GitHub tickets, review pull requests, and merge from testing into production

Jumbo Code Full-Stack Developer September 2025 – May 2026

Tufts University *Medford, MA*

- Built a dishware-lending web app for the Lexington Zero Waste nonprofit with a 13-person team (Next.js, React, TypeScript, Tailwind CSS) for 200+ potential users
- Wrote the RESTful API routes and Clerk authentication integration, backed by a PostgreSQL database via Prisma ORM; tested endpoints with Postman
- Worked in weekly-standup agile cycles with GitHub pull requests, code review, and a ticket board

Programming Languages Teaching Assistant January 2026 – Present

Tufts University Department of Computer Science *Medford, MA*

- Guide students through operational semantics, type systems, lambda calculus, and interpreter design in Standard ML and Scheme; evaluate parser, type-checker, and CPS assignments with detailed feedback

Game Projects

United Pigeon Services | *C#, Unity* October 2024 – December 2024

- Co-developed a 2D platformer in a team of 6; owned the pigeon's physics and controls: flight speed, glide time before falling, collision rules for which surfaces it lands on vs. flies through, and item pickup
- Fixed collision issues so the pigeon reliably stays on platforms; contributed to level design

Cow Catcher | *C#, Unity* October 2024

- Programmed the cow spawning algorithm and cow movement behavior relative to the player

Technical Skills

Languages: C#, C++, Java, Python, TypeScript/JavaScript, SQL, C, HTML/CSS, Standard ML, Scheme
Frameworks & Engines: Unity, React, Next.js, Node.js, Tailwind CSS, Pandas, Stable Baselines3, MuJoCo, Blender
Tools: Git, GitHub (PRs, code review, issue tracking), Postman, PostgreSQL, Prisma, Clerk, OBS Studio
Practices: Object-Oriented Design, REST APIs, Agile/Scrum, Code Review, IoT Systems, Reinforcement Learning